

Approaching Real-Time Fluorescence in RGB-Based Engines

Creating an artist friendly approach in Unreal Engine 5

1 Introduction

Offline and real-time renderers in computer graphics treat light as RGB values rather than wavelength behaviour. This results in issues when trying to represent spectral phenomena such as iridescence, thin-film interference, and fluorescence, as their behaviour is outside of RGB. Fluorescence involves energy conversion, high energy, ultraviolet (UV) light is absorbed and re-emitted as visible light, a lower energy light, and requires calculation to have plausible behavior

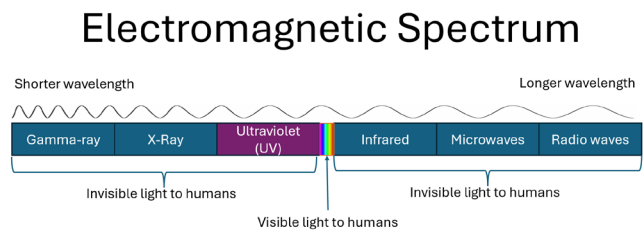


Fig. 1 Overview of Electromagnetic spectrum with UV light and Visible Light

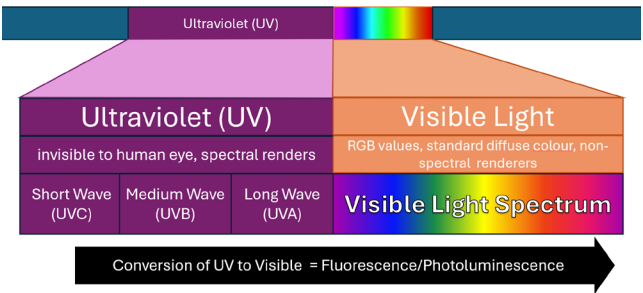


Fig. 2 Placement of Fluorescence in the Electromagnetic Spectrum

Currently, artists use emission maps to approximate static fluorescence. However, this method fails to maintain plausibility for all items that have fluorescence, even under studio light, nor for layered areas where the depth in this method is lost when under UV light, especially in real-time. This research aims to consider the proximity of light, ensuring that depth and detail are retained at an artist-controllable level without the need for specialized software.

2 Computer Graphics Foundation

Standard computer graphics are established by the Rendering Equation, which defines light leaving a surface as the combination of its own emission and the light it reflects (Kajiya 1986). Reflection is calculated through the microfacet theory, established by the Cook-Torrance (1982) model. These foundational models, however, assume that while light can be absorbed or scattered, the wavelength remains constant. This closed loop in modern Physically Based Rendering (PBR) models, including Unreal Engine’s shading model, lacks the ability to change wavelengths of light natively. Furthermore, in real-time rendering, performance requires significant approximation to maintain low render times, resulting in more behaviour losses, approximations, and pre-calculations.

As identified by Glassner (1995), fluorescence violates these closed loop and conversion assumptions. Most engines are monospectral, processing only one wavelength, visible wavelengths, fluorescence introduces re-radiation. In this process, UV light is absorbed and re-emitted in the visible spectrum, changing the wavelength from input to output, resulting in bispectral behaviour otherwise known as Stokes’ Shift. Since standard renderers calculate using RGB values, they ignore wavelengths outside the visible range and cannot natively approximate UV energy into RGB values (Akenine-Möller et al. 2018). To bridge this gap, Fichet et al. and Belcour et al. propose transitioning from full spectral to a Gaussian mixture model (Fichet et al. 2024, Belcour et al. 2025a). By utilizing parameters of shift, width, and yield, complex bispectral behaviour can be approximated to RGB renderers. Fichet, Belcour, and Barla (2024) used Unity, applying non-spectral (RGB) math to allow their model to prioritize behaviour over physical accuracy. Upon personal correspondence with the authors, this implementation isn’t designed to be art directed, it does not allow for artists to draw on the surface easily and the exposure to unfamiliar methods hinders artists (Belcour 2025, pers. comm.). While it allowed for easy mathematical control, it lacks pipeline implementation to update existing items to have fluorescence and to have a familiar way to add fluorescence at an artist level, like painting, without jumps in knowledge or workflow. Artists want to paint with colour and then adjust behaviours.

3 The Science of Fluorescence

3.1 Evolution of Fluorescent Rendering in Industry

The evolution of fluorescent rendering is defined by conflict between physical plausibility, render performance, and pipeline accessibility. Spectral renderers have achieved believable fluorescence (Fig. 3), but remain unintegrated into most animation pipelines. In 2022, Hua et al. (2022) advanced bispectral rendering through importance sampling and compressed storage (Fig. 5), though memory density makes real-time application impractical. Eclat Digital (2022) demonstrated mixed-lighting fluorescence in a proprietary context (Fig. 4), but is not well known in film.

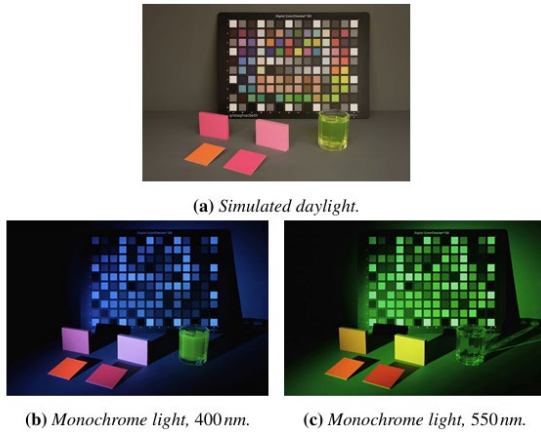


Fig. 3 Example fluorescence in spectral path tracer 2018

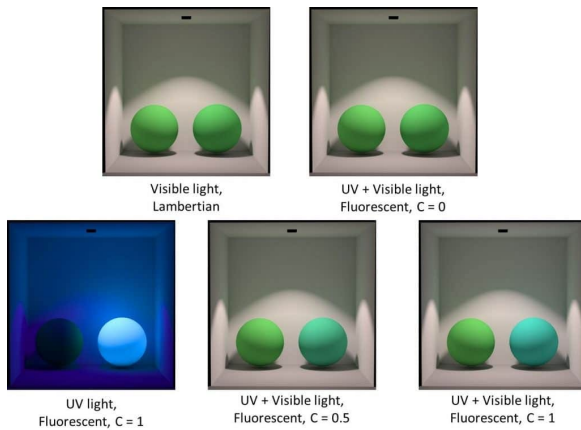


Fig. 3 Example Eclat digital (Eclat Digital 2022)

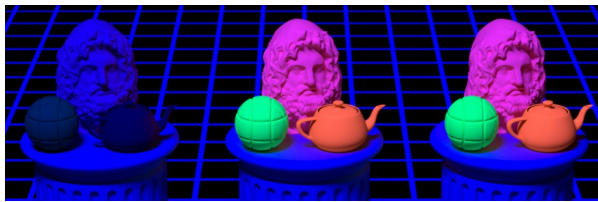


Fig. 4 Example render for storage and importance sampling (Hua et al. 2022)

By 2024, Fichet et al. (2024) produced a monospectral Unity approximation (Fig. 6) alongside their offline spectral paper, culminating in the 2025 Gaussian mixture

model (Fig. 7) designed for RGB and theoretically real-time rendering (Belcour et al. 2025a). However, GPU spectral approaches like in a 2021 implementation remain bottlenecked by Spectral Power Distribution (SPD), a complex curve and require specialized equipment, it cannot easily be generated by an artist (Fabre et al. 2021).



Fig. 6 Example render in Unity (Fichet et al. 2024)

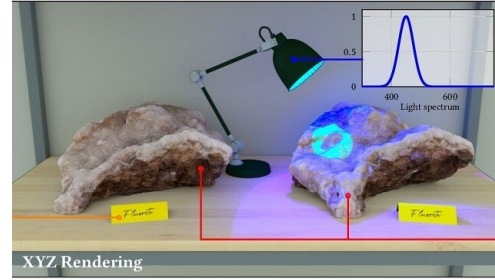


Fig. 7 Example render fluorescent model (Belcour et al. 2025a)

Gaussian mixture models offer variable control but demand scientific knowledge, and upon personal correspondence with Fichet (2025, pers. comm.), he confirmed colour clipping issues in implementation. Perez et al. (2025) addressed memory constraints via look up tables but target spectral reflections rather than surface behaviour. These trade-offs leave a consistent gap between plausibility, performance, and artist accessibility.

3.2 Fluorescence in Media

There are very few representations of Fluorescence due to the prior social perception of it being gimmiky, currently it is used for “rebellious” character depictions in animation such as Fig 8 and 9 (Padalino 2017). However, the known limitations as a result of a student project in Fig 11 brings the question of is the use of emission maps limiting or is it an intentional art direction. While these works have the advantage of render layers to help with plausibility, Fig 10 allows us to understand better the limitations applicably towards games.



Fig. 8 K/DA Popstars (Riot Games 2018)



Fig. 9 Love, Death, and Robots: Sonnie's Edge blacklight scenes (*Love, Death + Robots* 2019)

In Fig 10, this cleaning game only shows the dirt aspect being highlighted with one color and lacks the added depth of needed to remove stickers in game logic and in behavior. Similar conclusions were compared to my own photos in Fig 10 and 11 to distinguish what could be an oversight or a limitation of the easily accessible tools.

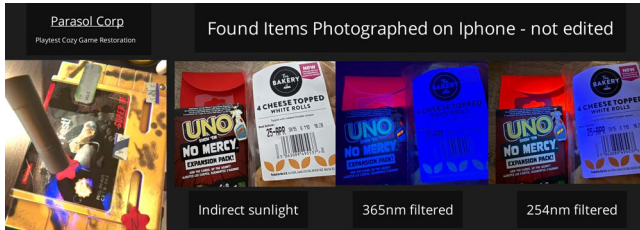


Fig. 10 Cozy Game Restoration vs personal found items (2026)

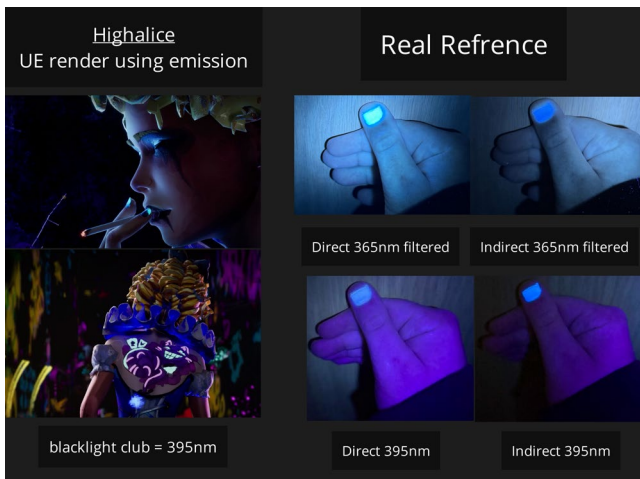


Fig. 11 HighAlice vs real reference (HighAlice 2026)

In HighAlice (2026), Unreal Engine's standard emissive workflow fails to capture the physical properties of fluorescence. The short's implementation lacks skin discoloration and fails to vary the emission strength based on light proximity or direction in Fig 11 resulting in a flat, non-reactive glow. The spectral accuracy is misaligned with the output intensity suggesting a short wave UV (365nm) as opposed to commercial black lights (395) that are long wave UV. Even in a professional implementation such as Fig 8, it also lacks a light proximity reaction. Additionally, both lack shadows on large amounts of fluorescent areas. While compositing of render layers can help shadow casting, real-time applications such as games require alternative implementations. By getting a more realistic setup, projects like these will not have to suffer from shadow casting issues yet not sacrifice artistic flexibility.

4 Prototyping

While the gaussian mixture model proposed by Belcour et al. (2025a) provides believability, its reliance on masks and abstract parameters hinders an artist's ability to intuitively paint fluorescent details like they would for texture maps. This research pivots toward direct manipulability, framing fluorescence as a dynamic gameplay mechanism rather than a static material property.

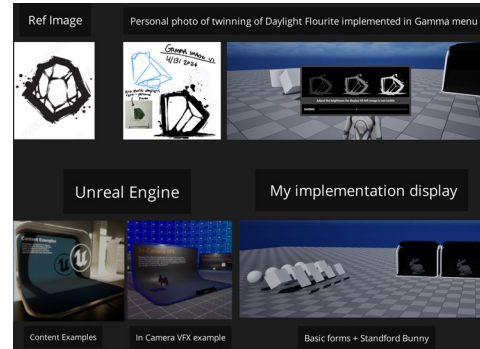


Fig. 12 Progress setup for basic handling in test level

In Fig 13 we see the issue of shadows and control over when the emission happens, alongside the glow overpowering the shadows. One implementation is to use Eye Adaptation to divide emission, however the glow is removed in Fig 14 despite presence of shadows.

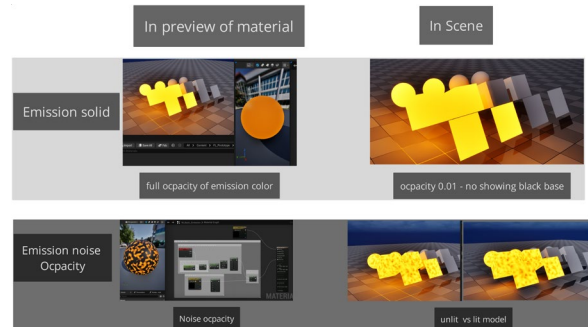


Fig. 13 Basic emission example in Unreal Engine with a lack of shadows, depth difficulty, flat emission

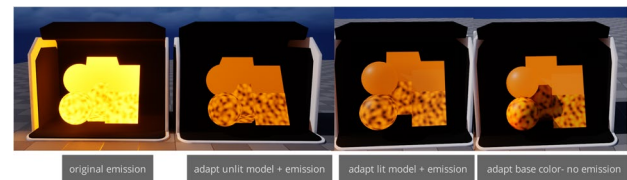


Fig. 14 Adding eye-adaptation to emission to bring shadows back in, depth is brought back

When initially handling going from an RGB value to controlling Gaussian parameters, one challenge is hue wrapping. If we simply remap it to 0-1, the wrappage results in both 0 and 1 being red as opposed to red and blue, requiring clipping.

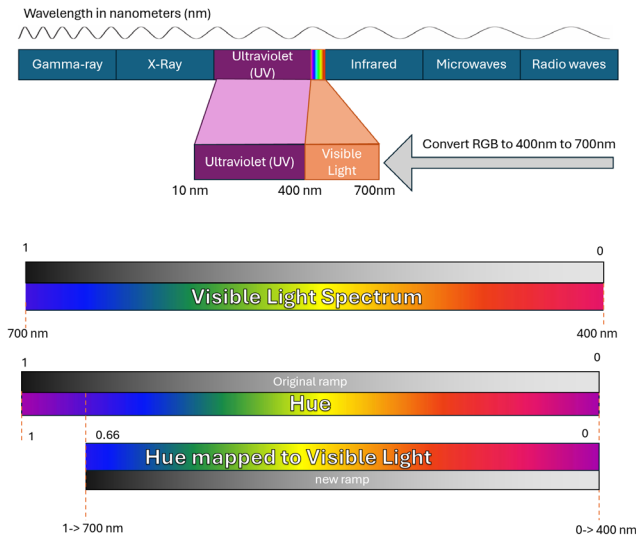


Fig. 15 Hue to Wavelength wrapping issue prior to the conversion of 0-1 to wavelength values of 400-700nm.

Handling upfront value issues in the conversion of texture to our 3 parameters in Fig 15 resulted in some further adjustments needed before a test was performed to see calculations with the native UE shader with emission channel for Fig 16. In the test, the base color matches but fails in shadow color being still overridden. Additionally, there is still color shifts that may need to happen, resulting in additional parameters to allow for control at this point.

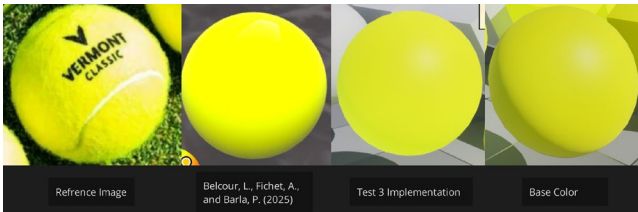


Fig. 16 Proof of concept test matching for handling hue to wavelength, processing, and output in comparison to the original Gaussian example (Belcour et al. 2025b)

When checking the rules of energy, Fig 16 the implementation of Stokes' Shift where it must have lower energy, later corrected. At this point, this is a single gaussian approximation as opposed to a gaussian mixture and my implementation uses a piecewise spectral ramp over CIE color matching, both compromises are faster but are less accurate.



Fig. 17 Unreal Engine test scene with the Stanford Bunny (Turk and Levoy 1994)

5 Conclusion

Compared to offline spectral models, this single Gaussian implementation prioritizes real-time scalability over spectral fidelity. This trade-off ensures the model remains optimized, friendly to an artist, while maintaining physical plausibility.

By mapping from a texture map to the re-radiation parameters of mean, gaussian width, and quantum yield, it prioritizes familiar use for artists. The deliberate shift from GMM and CIE integration to a piecewise spectral ramp significantly minimizes GPU overhead and LUT dependency, for optimization, but maintains rules such as Stokes' Shift. While currently optimized for the standard Unreal Engine shading model, further research into Unreal's Substrate system could extend this accuracy to complex volumes, liquids, and more complex materials.

Ultimately, this project demonstrates that physically inspired fluorescence is possible without dedicated spectral rendering. By balancing computational efficiency with artist agency, this proof of concept provides a scalable method for visualizing fluorescence, in its implementation potentially fostering broader public and scientific awareness of fluorescence in future games and animations. It is a model that allows for the expansion of individual lights triggering the effect because of optimization and is connected to a material parameter collection that can be connected to a light for the type of UV light to change its behaviour.



Fig. 18 Example proof of concept using noise as the texture input using the Stanford Bunny (Turk and Levoy 1994)

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Appendix

1A Notable Properties in Photoluminescence

1A.1 Important Terms and Behaviors

Electromagnetic Spectrum: The entire range of wavelengths, measured in nanometers, spanning from high-energy gamma rays to low-energy radio waves.

Nanometers (nm): 10^{-9} meters, the standard unit of measurement for wavelengths within the electromagnetic spectrum.

Wavelength: The distance between two points on a wave (e.g., peak to peak).

Visible Light: The narrow band within the electromagnetic spectrum (approximately 400–700nm) that is detectable by the human eye.

Ultraviolet (UV) Light: Electromagnetic radiation with higher energy than visible light (below 400nm) that is invisible to the human eye.

Photoluminescence: An umbrella term for substances that absorb photons and re-emit them.

Fluorescence: A type of photoluminescence where the material absorbs high energy electromagnetic radiation, UV light, and near instantaneously re-radiates it as a lower energy at visible light.

Re-Radiation: The behaviour of shifting from an invisible spectrum to a visible spectrum resulting in a bispectral behaviour

Bispectral: Systems capable of calculating light that enters at one wavelength and exits as another.

Monospectral: Systems that process light as a single wavelength or a fixed 3-channel (RGB) value.

Stoke's Shift: The specific energy conversion where high-energy UV light is absorbed and re-emitted as lower-energy visible light.

Quantum Yield: The ratio of photons emitted to photons absorbed, which determines the brightness of the fluorescence.

Spectral Power Distribution (SPD): A complex data set representing the specific power of a light source across every individual wavelength in the spectrum.

Spectral Renderers: High-end simulation engines (e.g., Maxwell, Weta's Manuka) that use wavelength-based math to be more physically precise but computationally slow.

Non-Spectral Renderers: Standard engines (like Unreal or Unity) that use RGB (red, green, blue) math but cannot natively calculate wavelength shifts.

Gaussian Mixture Model (GMM): A mathematical approach that approximates a complex SPD curve by combining several simpler Gaussian (bell-shaped) curves.

Look-Up Table (LUT): A pre-calculated set of values stored to speed up complex mathematical operations, allowing for higher performance in real-time rendering.

1A.2 Crystal Classics

A special thank you is extended to UK Mining Ventures and Crystal Classics for their assistance in my investigation of various fluorescence properties. Their support in providing opportunities to document specimens of UK daylight fluorescent fluorite, the mineral that "fluorescence" got its name from. I gained insight into the most accepted theory of Rare Earth Metals (REEs) acting as activators for this strain. Further first-hand documentation elevated my understanding of fluorescence beyond recognizable lights shown but also more passive exposure to UV light.



Fig. A1 Selection of daylight fluorescent fluorite from a variety of pockets from Crystal Classics' specimens



Fig. A2 Difference of color and other properties like iridescence that can be found in UK daylight fluorescent fluorite

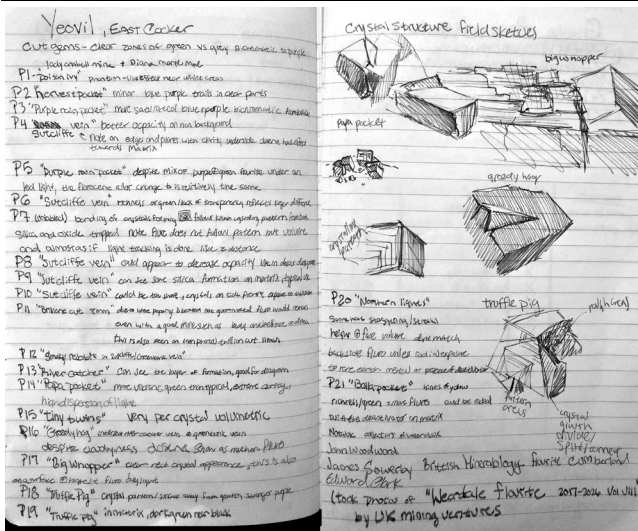


Fig. A3 Notes from photos collected and drawing studies

1A.3 Jurassic Coast

Given my prior knowledge, I wanted to test a hypothesis along Monmouth Beach and Lyme Regis Beach given the presence of fossils and the mineral diversity there was an chance to have phosphorescent behaviors as well. To my knowledge, presence of phosphorescent minerals was not recorded in the area, however its common given my background for new mines to have discoveries as well given someone experiments with UV wavelengths against minerals. As a result of the trip my hypothesis driven by the presence of fossils resulted in both phosphorescent and fluorescent behaviors that went undocumented till now. Below are some highlights of samples collected of what one may find along the Jurassic Coast. To me, this is a discovery that has gone unnoticed until equipment became lower power and more accessible to allow for an individual discovery despite being a tourist area, you must be curious enough to shine a light and see. The more people do this, the more we can understand as to why these behaviors occur. As more people are exposed to these properties, the more understanding we can have about our world.



Fig. A4 Selected one may find along the Jurassic Coast

While more phosphorescence can be seen on video, on fossils like chionids, the right most image in Fig A4, and more have a pale yellow that either phosphoresces light blue/green or slightly more saturated yellow which shows the mineral replacement. The banding identifies chemical shifts in water as the mineral forms for the left most one in Fig A4.

The small field study I performed demonstrates that as UV equipment becomes more accessible, the barrier to

individual scientific discovery is lowered, and using animation to further illustrate these discoveries can be used to encourage more identities to better understand our environment's behaviour, its past, and look at it holistically. Even in the most studied environments, filled with tourism, there will always be more to discover and share with people.

1A.4 Personal Collection

Several of my own minerals have fluorescent properties. These real-life examples in the strength of the lightbulbs aided in the tests I performed throughout this project. In Fig A5, if a mineral reacts to one light or wavelength does not determine if it reacts to higher or lower ones. How direct a light is, clarity of a light all changes the interaction. This difference is one of the reasons why I wanted to highlight interaction differences as an important for an artist to control.

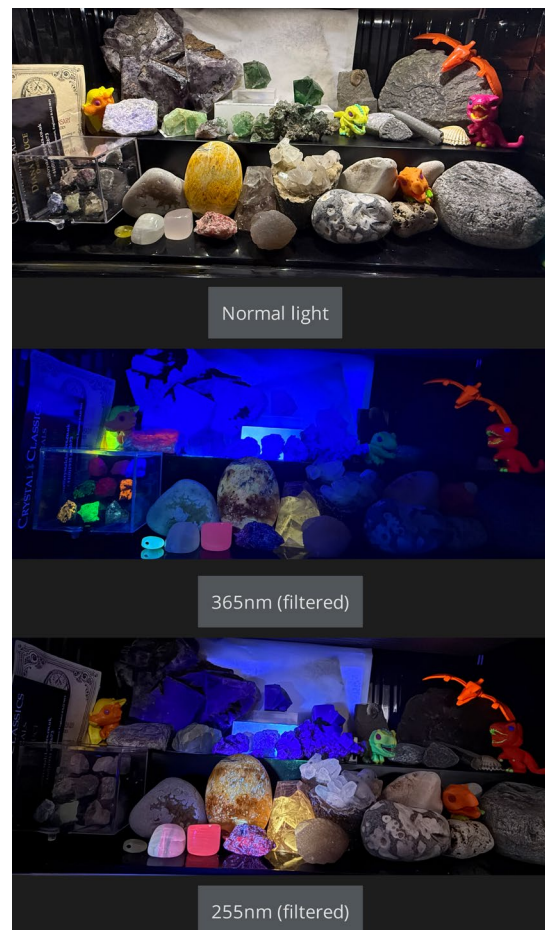


Fig. A5 Light reaction for minerals vs synthetic

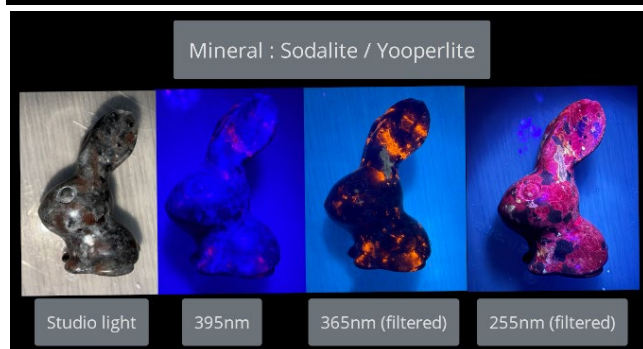


Fig. A6 Main reference for display interaction

2A Looking Forward in Implementation

Through designing a way for an artist to interact with the fluorescence that is familiar to them, it allows for these expansions to happen easily after a prototype has been established. I truly believe that as visibility for these properties increases, as the technology to see this phenomenon becomes more accessible, more scientific discoveries will be made across disciplines. The following is a few ways I can see a real time system being developed differently or the current one expanded upon.

Through the values in *The World of Fluorescent Minerals*, Schneider gave approximate Kelvin, adjusted in Table 1 according to other resources and further estimations to fill in scenarios, that can aid in potentially other implementations in determining states. Something to note though is in both of Schneider's books, *The World of Fluorescent Minerals* and *Collecting Fluorescent Minerals*, alongside my trials in photographing fluorescence, yellow fluorescence requires to be altered as it shifts green upon photographing without any colour correction, even still brightness levels from my own personal photographs were higher than what the human eye could see as opposed to the camera. This needs to be considered when viewing resources such as Schneider's books, Fluomin.org, Mindat.org, or hobbyist's records for an intentional implementation. Like many spectral behaviors, a physical object is preferred if the system is not a system based on scientific data to create plausibility, especially in the human eye vs a camera.

Table 1 Different UV settings, Approximate Kelvin

*Please note artistic liberties in visualizing lights, some of these are not measurements

Kind	Wavelength	Kelvin Adjusted	Kelvin (Schneider 2006)
Daylight	400-700nm	5600K	5600K
Overcast	~470nm	7000K	7000K
Candlelight	~650nm	1900K	1900K
Studio Light (Tungsten)	~600nm	3200K	N/A
Blue Sky	400-450nm	20,000K	20000K
Long Wave	395-365nm	10,000K	(Higher than SW, blue)
Medium Wave	280-315nm	15,000K	N/A
Short Wave	254-255nm	25,000K	>20,000K

Furthermore, foundational databases like the BISBRDF lookup tables developed by Hullin et al. could be modernized (Hullin et al. 2010). By adding measurements from diverse mineral databases, and others created, these architectures could be updated to support spectral behaviours in the future.

Other properties like thermoluminescence or triboluminescence present greater implementation challenges as they are not light-based behaviours. However, by grouping photoluminescent traits by their behavioural triggers, as seen in *The World of Fluorescent Minerals*, current models designed for fluorescents can be expanded (Schneider, 2006). By adjusting logic regarding exposure duration, the current work of Fichet, Belcour, and Barla can be expanded to simulate other related properties. For example, phosphorescence involves delayed emission; by adding a temporal decay, properties can be implemented like in my primary research. Similarly, tenebrescence involves a colour shift based on UV exposure, time exposed to and time after to shift the base colour. Flashes of red in phosphorescence, different phosphorescence colour based on type of light exposed to could all increase an accessible visibility of mineral or other scientific data.

E-mail: s5501023@bournemouth.ac.uk